Mehrdad Shafiei Dizaji, Ph.D.

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EXPERIENCE

Postdoctoral Research Associate Jun 2023 – Present

Turner Fairbank Highway Research Center, FHWA *McLean, VA*

- Applying Physics-Informed Neural Networks to predict NDE data for highway infrastructure assessment applications.
- Leading ongoing research on NDE data analysis using artificial intelligence through the NDEChat platform.

PROJECTS

Load Rating Strategies for Bridges with Missing or As-Built Information 2014 – 2018

- Developed a 3D Finite Element Model-driven methodology for load rating of bridges with missing or as-built information.
- Designed instrumentation and conducted full-scale field load testing of 4 highway bridges, analyzing data from physical sensors.
- Collaborated in writing the final project report for VDOT; sponsored by VTRC.

Evaluation of Five Spans of the Hampton Roads Bridge Tunnel 2016 – 2016

- Collaborated with 5 UVa researchers and 8 VDOT employees on field load testing of the Hampton Roads Bridge Tunnel.
- Analyzed physical sensor data and performed vision-based structural assessment in a complex testing environment.

Evaluation of Mecklenburg Bridge Following Repair Using 3D Digital Image Correlation 2018 – 2018

- Conducted vision-based structural assessment and applied a hybrid computer vision approach for holistic damage detection.
- Investigated the behavior of a bridge beam previously replaced due to crippling failure and web buckling.

Continued Monitoring of the Varina-Enon Bridge Response to Special Permit Loads 2019 – 2020

- Implemented 3D Digital Image Correlation as a non-invasive, full-field measurement method to obtain in-plane and out-of-plane strains for structural identification.

Preserving Coastal Infrastructure through Image-Based Structural Health Monitoring 2016 – 2020

- Developed an image-based structural characterization technique to assess in-service infrastructure components non-invasively.
- Utilized high-resolution 3D laser scanning (HR3DLS) to measure surface defect geometry and map defects into numerical simulation.
- Applied full-field measurements from 3D digital image correlation (3D-DIC) for refined structural identification (St-ID).

SKILLS

Structural and FEA Software: ABAQUS, ANSYS, SAP2000, LARSA 4D, CSIBridge, PGSuper, ARTeMIS

Diagnostics Expertise: Non-Destructive Evaluation (NDE), 3D Digital Image Correlation (3D-DIC), Physical Sensor-Based Diagnostics, Image-Based Diagnostics

Programming and ML: Python, MATLAB, Fortran, PyTorch, TensorFlow, Scikit-learn

Other Tools: AutoCAD, LATEX, Microsoft Office, Ubuntu

CERTIFICATIONS

- MatchID Digital Image Correlation Technique, Drexel University, Philadelphia — Jan 2018

EDUCATION

University of Virginia

Ph.D. in Civil and Environmental Engineering. GPA: 4.00. *May 2020*

Relevant Coursework: Smart Structures, Continuum Mechanics, Non-Destructive Evaluation, Finite Element Analysis, Machine Learning, Mechanics of Composite Materials

Sharif University of Technology

M.Sc. in Structural Engineering. GPA: 3.80. *June 2008*

University of Tabriz

B.Sc. in Civil Engineering. GPA: 3.67. *June 2005*